Complex Analysis Final Examination

2025–2026 First Semester

Standard Study Version

1. Let $G, \Omega \subset \mathbb{C}$ be domains, let $u : G \to \mathbb{R}$ be harmonic, and let $f : \Omega \to G$ be holomorphic. Using $\Delta = 4\partial\bar{\partial}$, prove that

$$\Delta(u \circ f) = ((\Delta u) \circ f)\, |f'|^2.$$

In particular, $u \circ f$ is harmonic on Ω.

2. Use contour integration to evaluate

$$\int_0^\infty \frac{dx}{x^a(1+x)}, \qquad 0 < a < 1.$$

3. Let $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$.

(a) Suppose f is holomorphic on $\mathbb{D}$, $|f(z)| \le M$ on $\mathbb{D}$, where $M > 0$, and $a_1, \dots, a_n \in \mathbb{D}$ are zeros of f, listed with multiplicities not exceeding their orders. Prove that

$$|f(z)| \le M \prod_{j=1}^{n} \left| \frac{z - a_j}{1 - \overline{a_j} z} \right|, \qquad z \in \mathbb{D}.$$

Hint. Use the method of proof of Schwarz's lemma.

(b) A finite Blaschke product is a function of the form

$$B(z) = e^{i\theta} \prod_{j=1}^{n} \frac{z - a_j}{1 - \overline{a_j} z}, \qquad a_j \in \mathbb{D}, \quad \theta \in \mathbb{R},$$

where $n \ge 0$ and an empty product is interpreted as 1. Suppose f is holomorphic on $\mathbb{D}$, continuous on $\overline{\mathbb{D}}$, and $|f(z)| = 1$ for $|z| = 1$. Prove that f is a finite Blaschke product.

4. Let f be meromorphic on $\mathbb{D}^* = \{z \in \mathbb{C} : 0 < |z| < 1\}$. Suppose f has a sequence of distinct poles $z_j \to 0$. Prove that the set of its finite values

$$\{f(z) : z \in \mathbb{D}^*,\ z \text{ is not a pole of } f\}$$

is dense in $\mathbb{C}$. *Hint.* Use the method of proof of the Casorati–Weierstrass theorem.

5. Let $\Omega \subset \mathbb{C}$ be a domain with $0 \in \partial\Omega$ and $[0,1] \subset \mathbb{C} \setminus \Omega$. Construct a local subharmonic barrier at 0. More precisely, find $r \in (0,1)$ and a real-valued function v, continuous on $\overline{\Omega} \cap \overline{D(0,r)}$ and subharmonic on $\Omega \cap D(0,r)$, such that

$$v(0) = 0, \qquad v(z) < 0 \quad \text{for every } z \in (\overline{\Omega} \cap \overline{D(0,r)}) \setminus \{0\}.$$